

Employee Attrition Prediction –Project Summary

Employee Attrition Prediction Using Machine Learning

Project Summary

The purpose of this project was to identify employees who are more likely to leave the company by analyzing employee information such as department, job role, monthly income, work-life balance, years at the company, and other workplace factors.

The analysis found that employee attrition is higher in the **Sales department**, especially among **Sales Representatives**. Employees with lower monthly income, poor work-life balance, frequent business travel, overtime, and fewer years at the company were more likely to leave the organization.

A predictive model was developed to help identify employees who may be at risk of leaving. Among the models tested, **Logistic Regression** provided the best overall performance for this dataset and was selected as the final model.

Based on the findings, the company should focus on improving employee retention by supporting high-risk departments and job roles, reducing excessive overtime, improving work-life balance, providing career growth opportunities, and regularly collecting employee feedback. These actions can help improve employee satisfaction and reduce attrition.

This project demonstrates how data analysis can support Human Resources in making informed decisions and developing effective employee retention strategy